

AHJIN STUDIOS PRODUCTION

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Lights, Camera ... Data!

*Analysis By: The
Standard Deviants*





Business Goal

- ☐ To maximize the commercial success of Ahjin Studios' film projects by using industry data to make informed choices about genres, budgets, release timing, and marketing strategy.
- ☐ To estimate the budget costs of creating a film and the ROI.

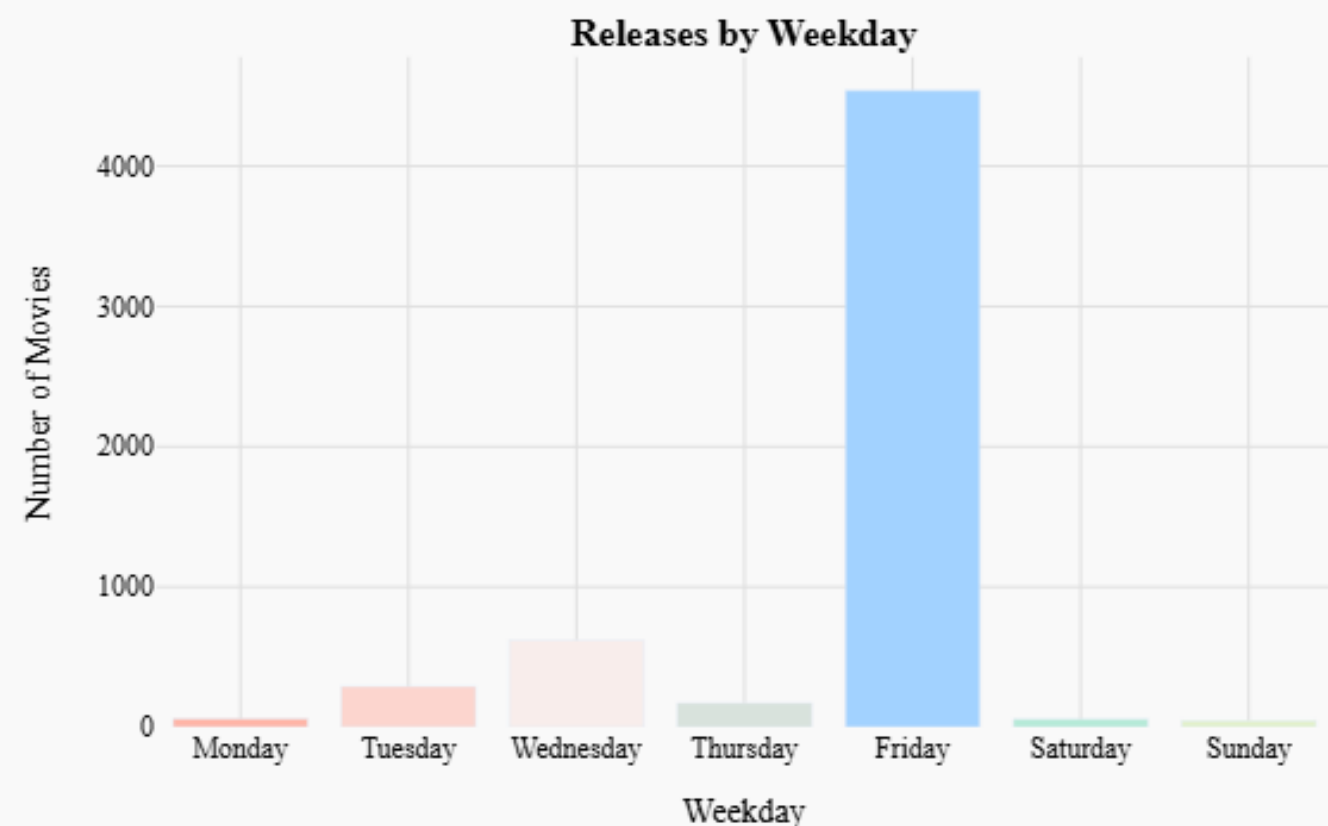
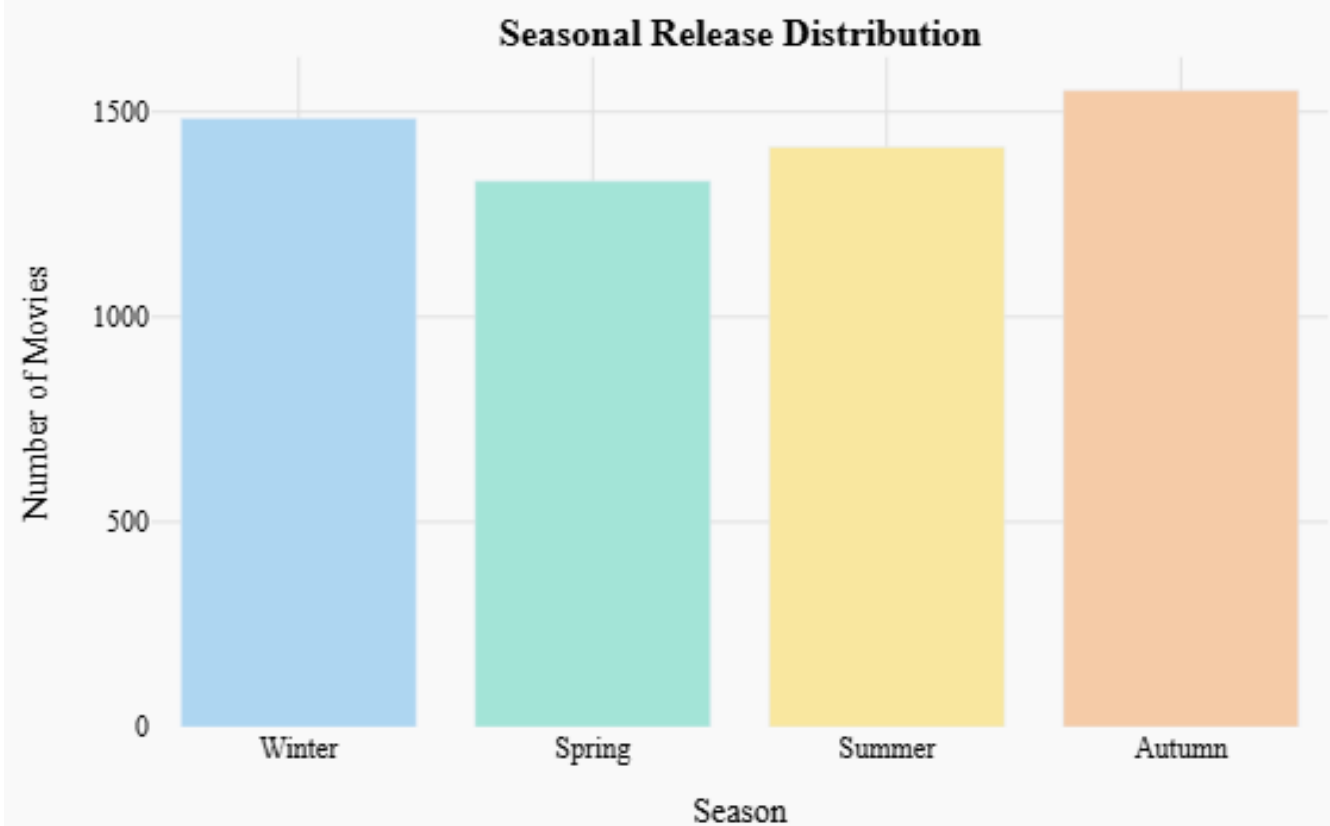
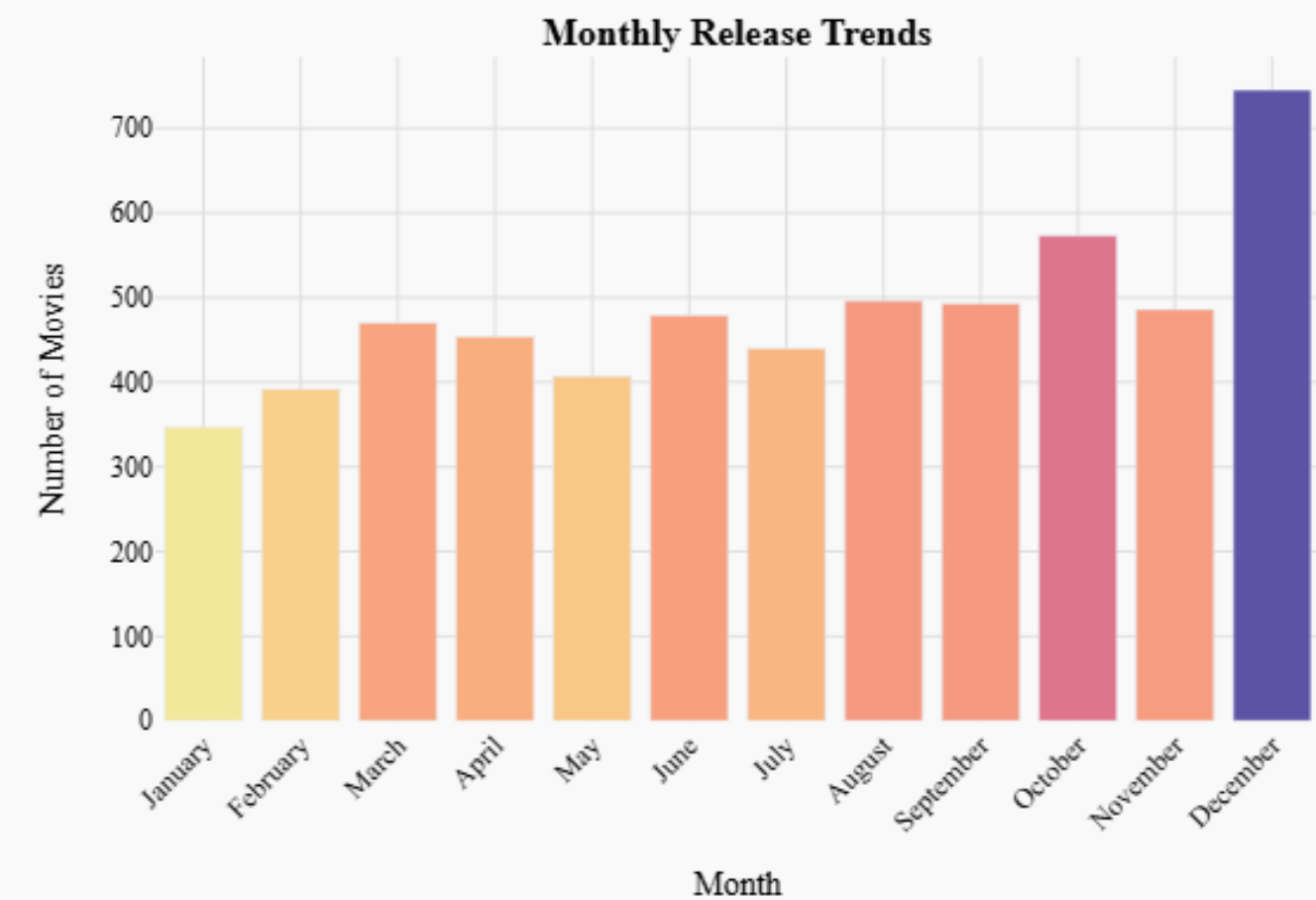
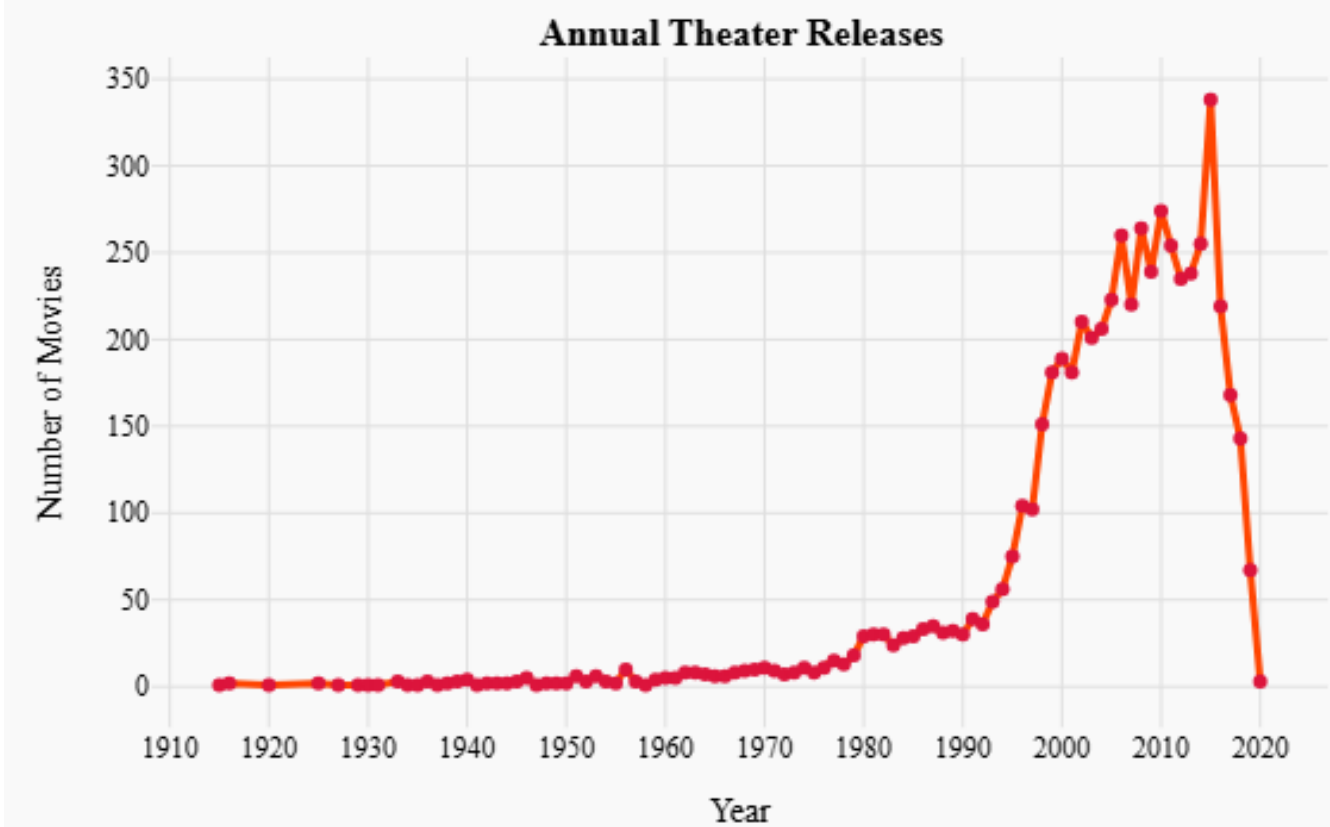
About the Project

- ✧ Ahjin Studios is a fresh face in film, using market insights to analyze recent box office trends and translate key findings into **actionable, data-driven recommendations**, so as to smartly choose and shape movies for success in a tough, ever-changing industry.



WHEN IS THE BEST
TIME TO RELEASE A
MOVIE ?

Movie Release Trends Overview by The Numbers



Observation

Key Trends in Film Releases

Decade Trends

- ❑ Releases surged from the 1980s onward, peaking in the 2000s (>10,000 films).
- ❑ The 2010s saw a slight decline (7,787 films), possibly due to streaming shifts.

Monthly Patterns

- ❑ **Peak Months:** December (3,154), November (3,023), January (2,537) (holiday/awards season).
- ❑ **Summer Blockbusters:** June, July, and August remain strong.
- ❑ **Lowest Month:** February (1,569 films) due to post-awards slump.

Seasonal Trends

- ❑ **Highest Releases:** Autumn (7,643) and Winter (7,260) dominate (awards/holiday focus).
- ❑ **Lowest Releases:** Spring (5,591) has weaker commercial appeal.

Weekday Releases

- ❑ **Friday** dominates (20,048 films) for weekend box office success.
- ❑ **Midweek (Wednesday: 3,402)** likely for limited/festival releases.
- ❑ **Weekends (Sat/Sun)** are least popular for launches.

Monthly Film Releases: Theater vs DVD by Box Office Tier



Observation

Theater Releases by Box Office Tier

- **Blockbusters (\$100M+)**: Appear consistently but **peak in June, November and December**, aligning with holiday season strategy.
- **Mid-tier (\$10M–\$50M)** and **Low-tier (\$1M–\$10M)** releases hold steady throughout the year, **spiking in October**.
- **High-grossing (\$50M–\$100M)** show strongest in **December**, aligning with awards and holiday family crowds
- **Flops (<\$1M)** surprisingly cluster around **April to July**, perhaps indie or experimental releases.

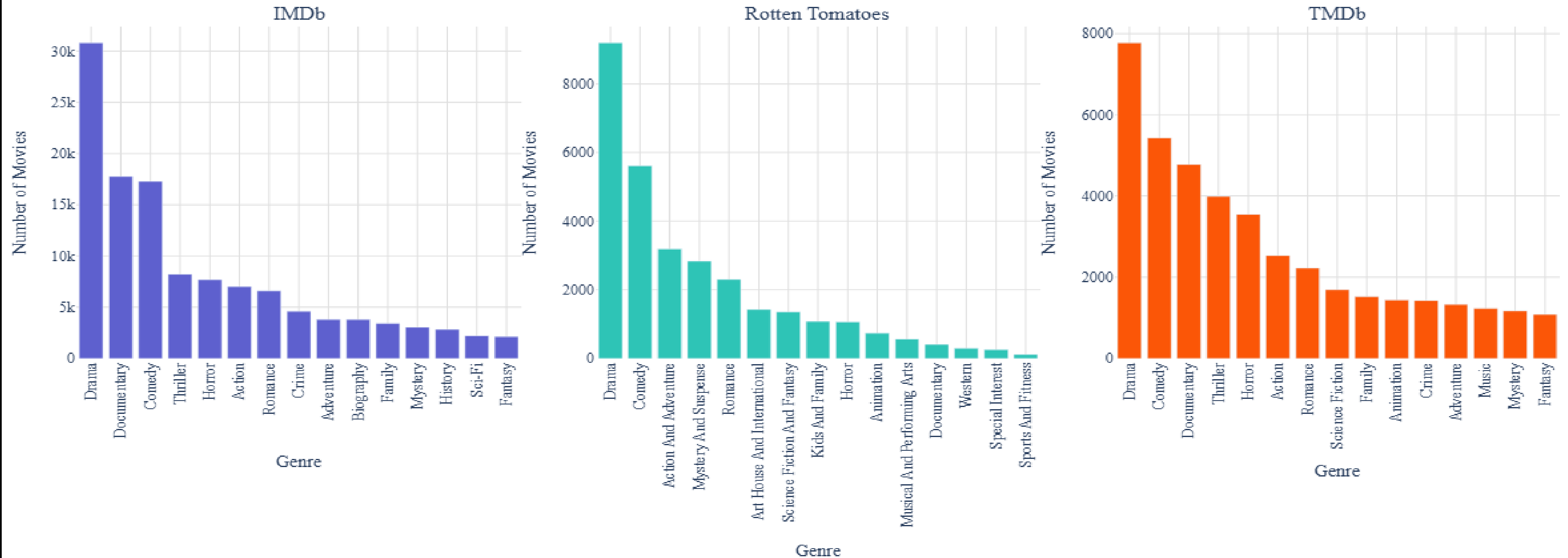
DVD Releases by Box Office Tier

- **Blockbusters and High Grossers** prefer **November–December DVDs**, banking on holiday shoppers.
- **Flops** get DVD drops early in the year - **January to March**- possibly as cleanup from the previous year.
- **Mid & Low grossers** maintain consistent DVD drops, rarely spiking, indicating stable long-tail value.



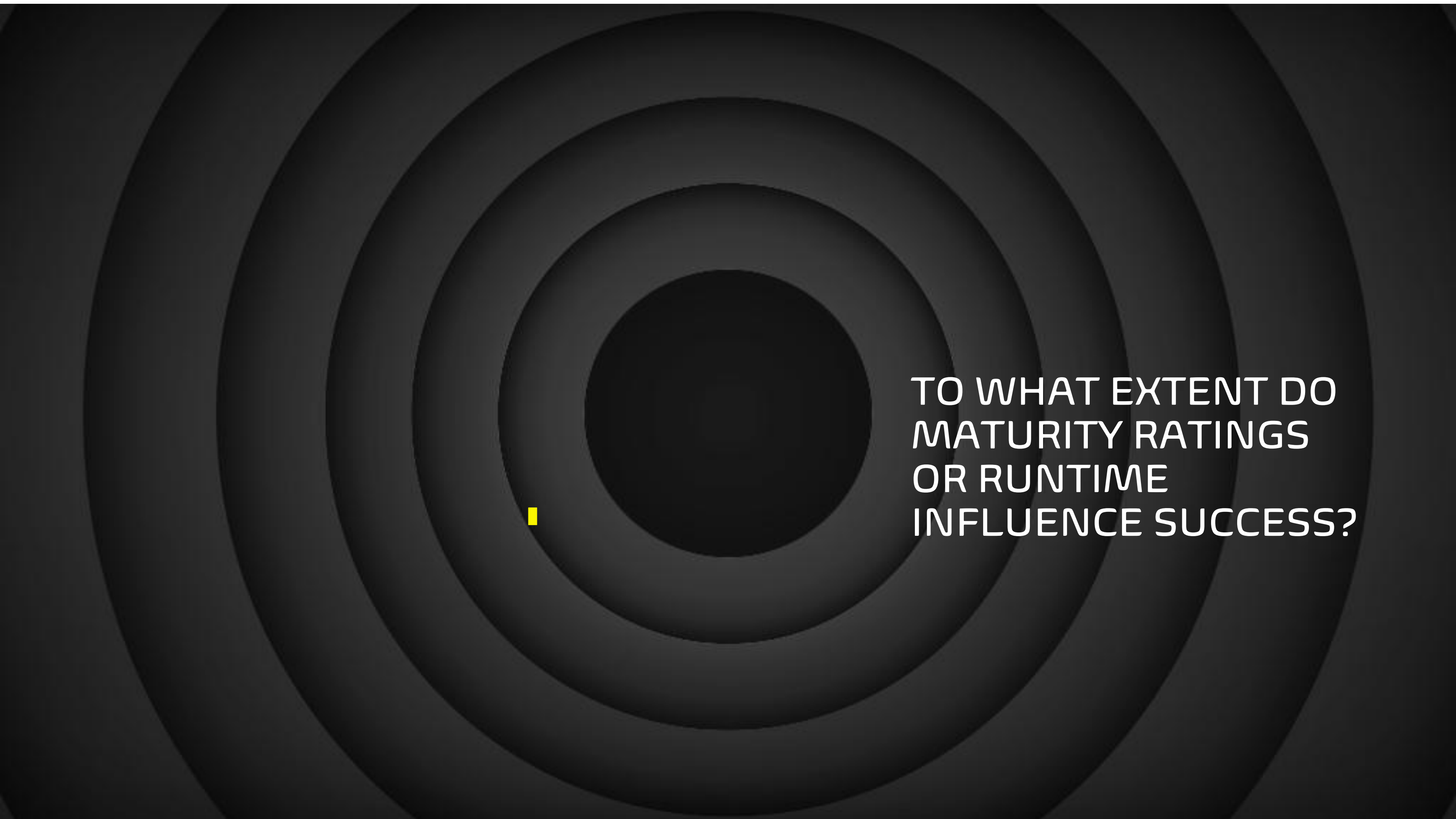
WHAT ARE THE TOP GENRES
ACROSS DIFFERENT
PLATMORMS?

Top 15 Genres by Platform: IMDb vs Rotten Tomatoes vs TMDb



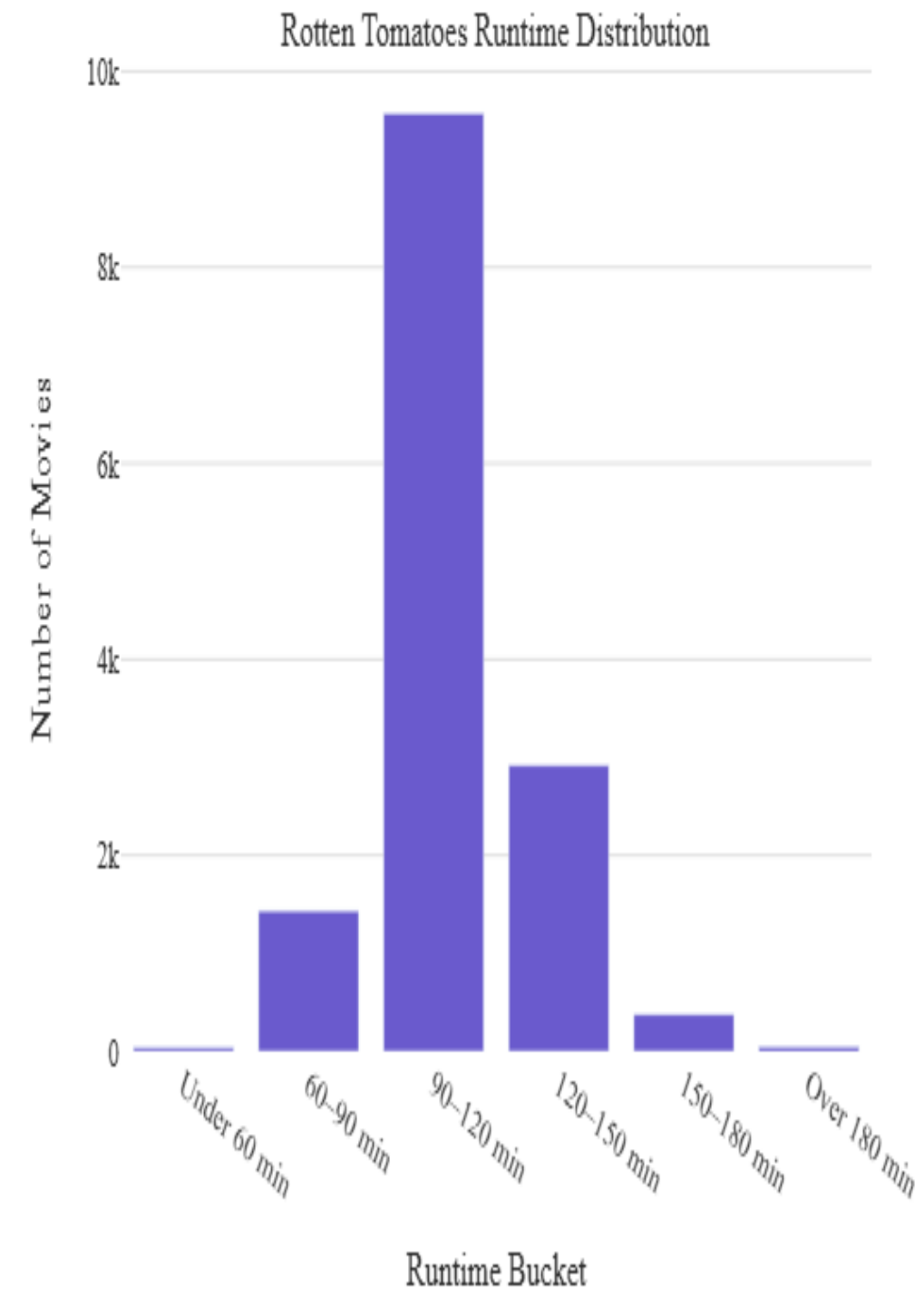
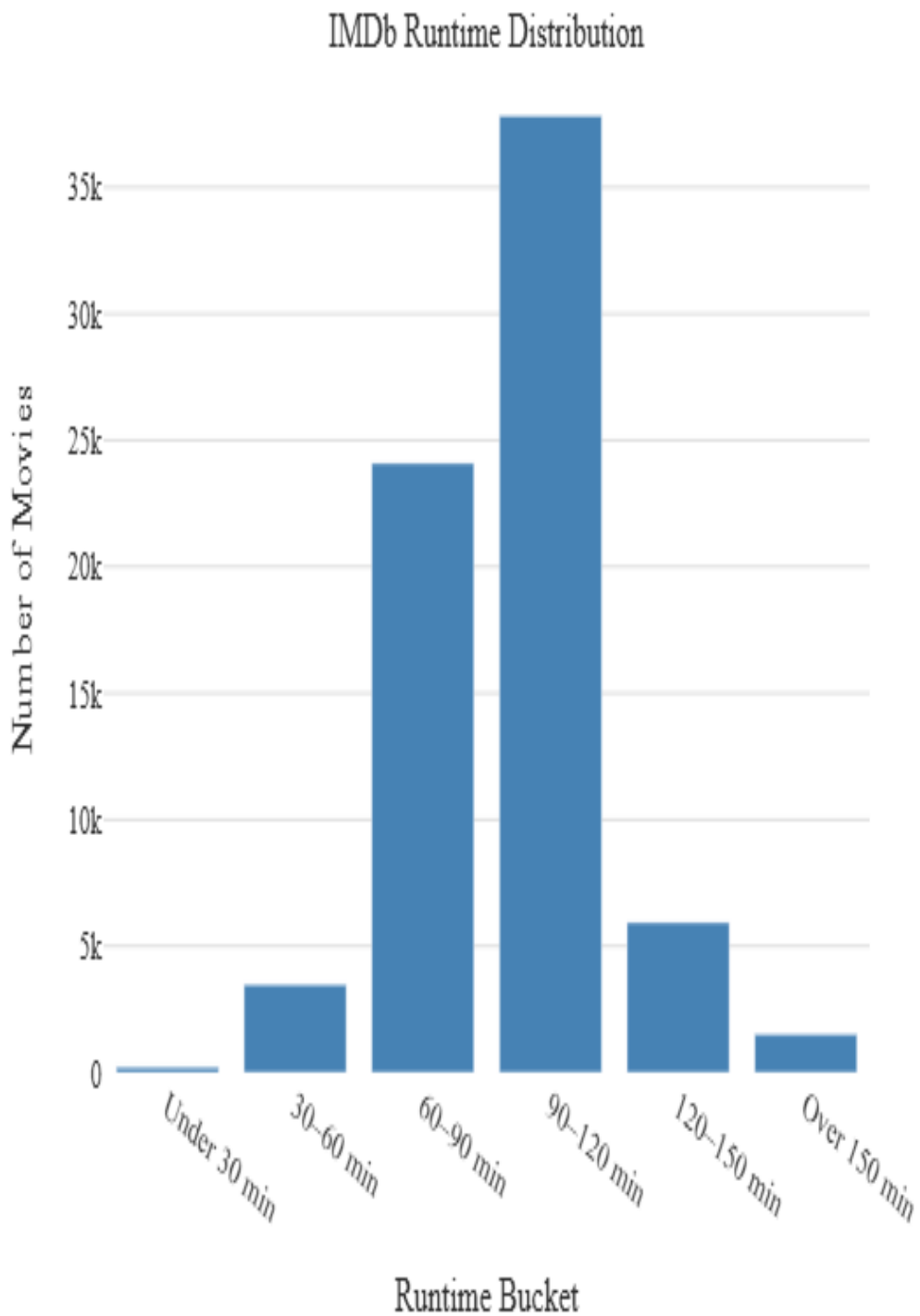
Top 15 Genres: Platform Comparison

- **IMDb:** Clean, standardized genres. *Drama* dominates, followed by *Documentary* and *Comedy*. Structured taxonomy.



TO WHAT EXTENT DO
MATURITY RATINGS
OR RUNTIME
INFLUENCE SUCCESS?

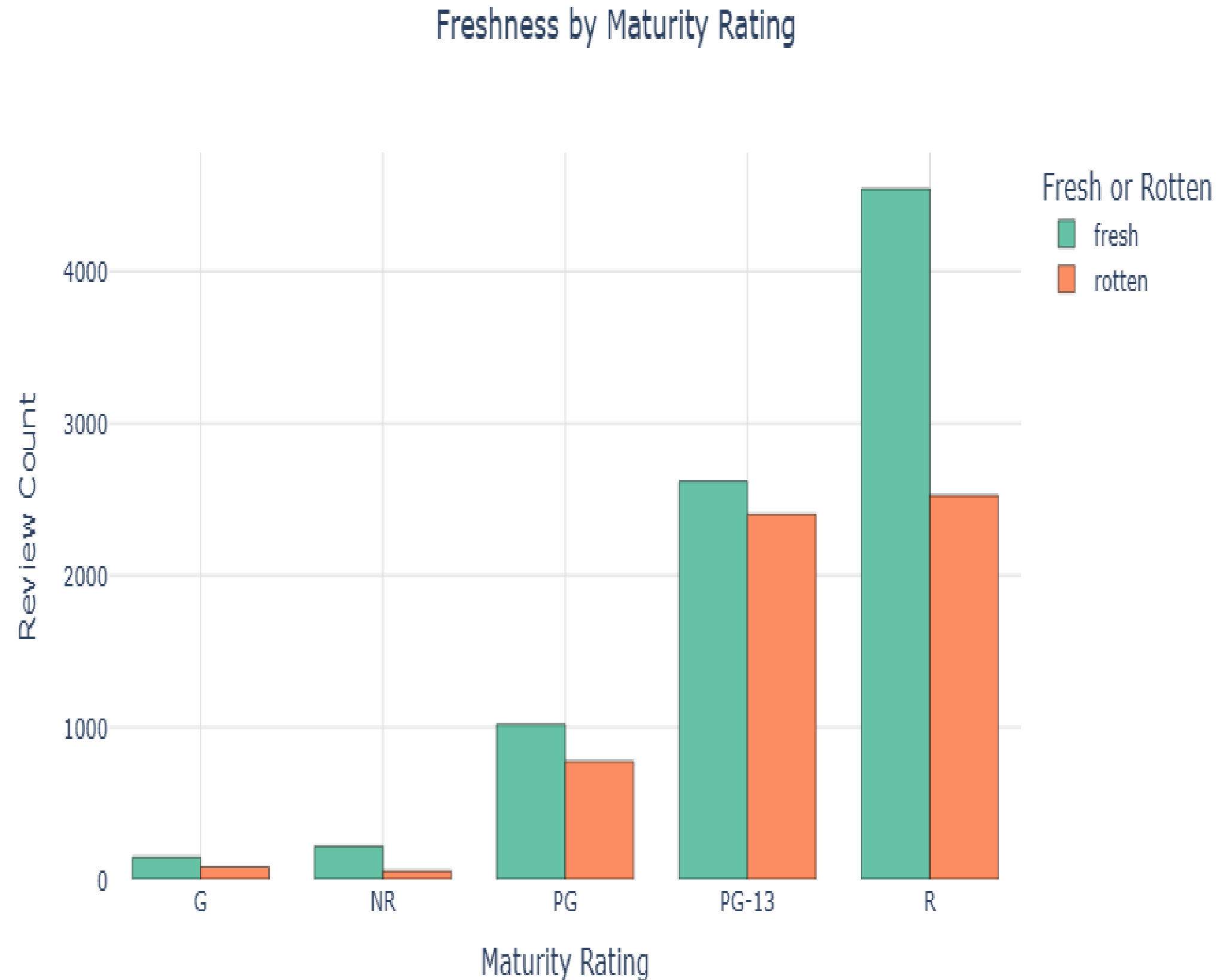
Runtime Distribution Comparison: IMDb vs Rotten Tomatoes



- Across both platforms, **90–120 minutes is the industry sweet spot.**
- IMDb: 37,832 films (largest share)
- Rotten Tomatoes: 18,073 films (also dominant)
- **60–90 minute films are highly represented on IMDb** (24,099 entries), but much less on Rotten Tomatoes—suggesting they’re common, yet not often critically celebrated.
- Shorter formats (under 60 minutes) and ultra-long features (over 150 mins) are **niche segments**—they exist, but rarely drive mainstream buzz or critical acclaim.
- IMDb uniquely records **a significant volume of shorts** (under 30 mins and 30–60 mins), offering insight into experimental and indie trends, but less applicable to theatrical strategy.

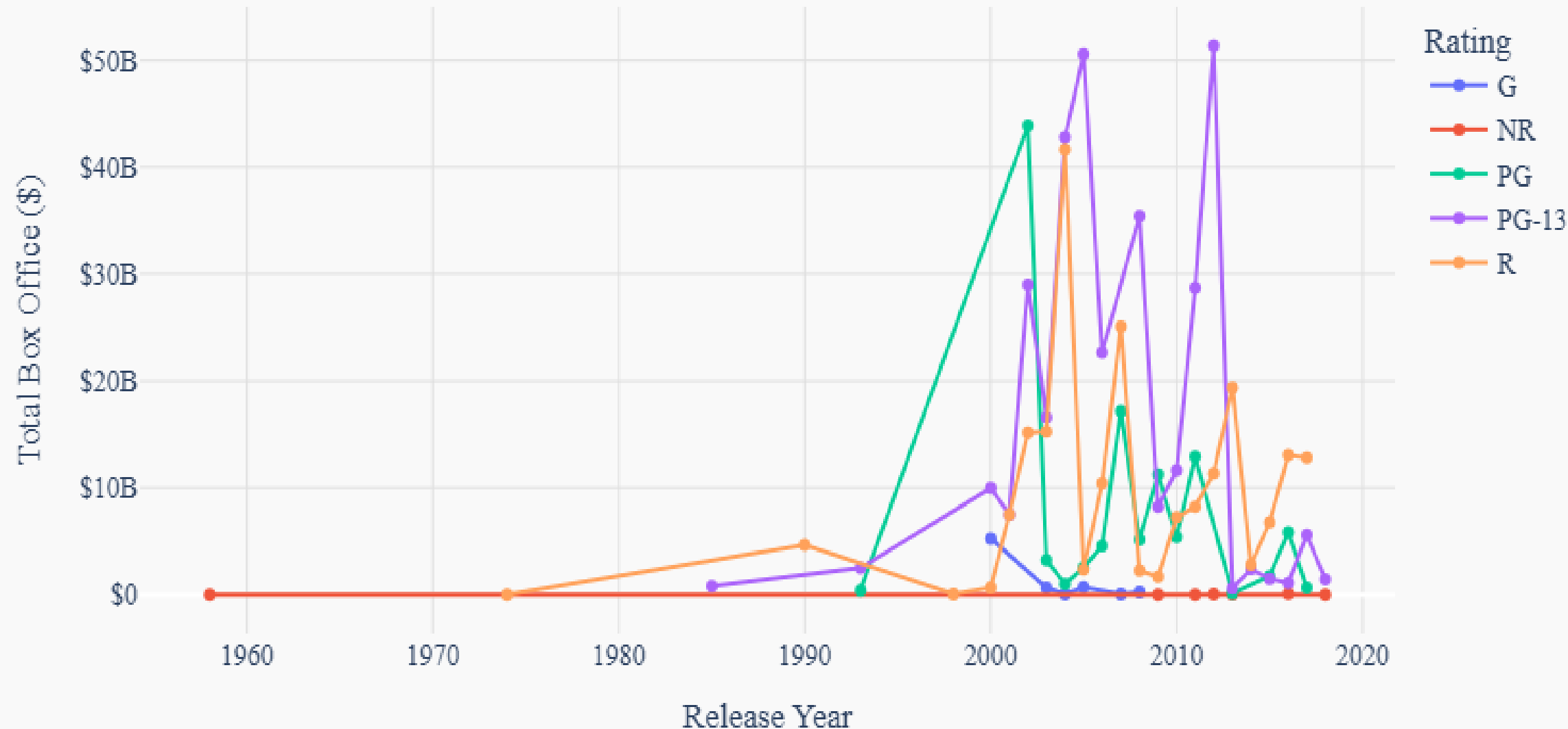
INSIGHT:

- NR is critics' favorite (78.6% Fresh)
- PG-13 floods the market but splits critics.(54.5%)
- R balances mass and quality(63.7% Fresh)
- G/PG = safe but less impactful.(74% & 64% Fresh respectively)



TO WHAT EXTENT DO MATURITY RATINGS OR RUNTIME INFLUENCE SUCCESS?

Box Office Revenue by Maturity Rating (Over Time)



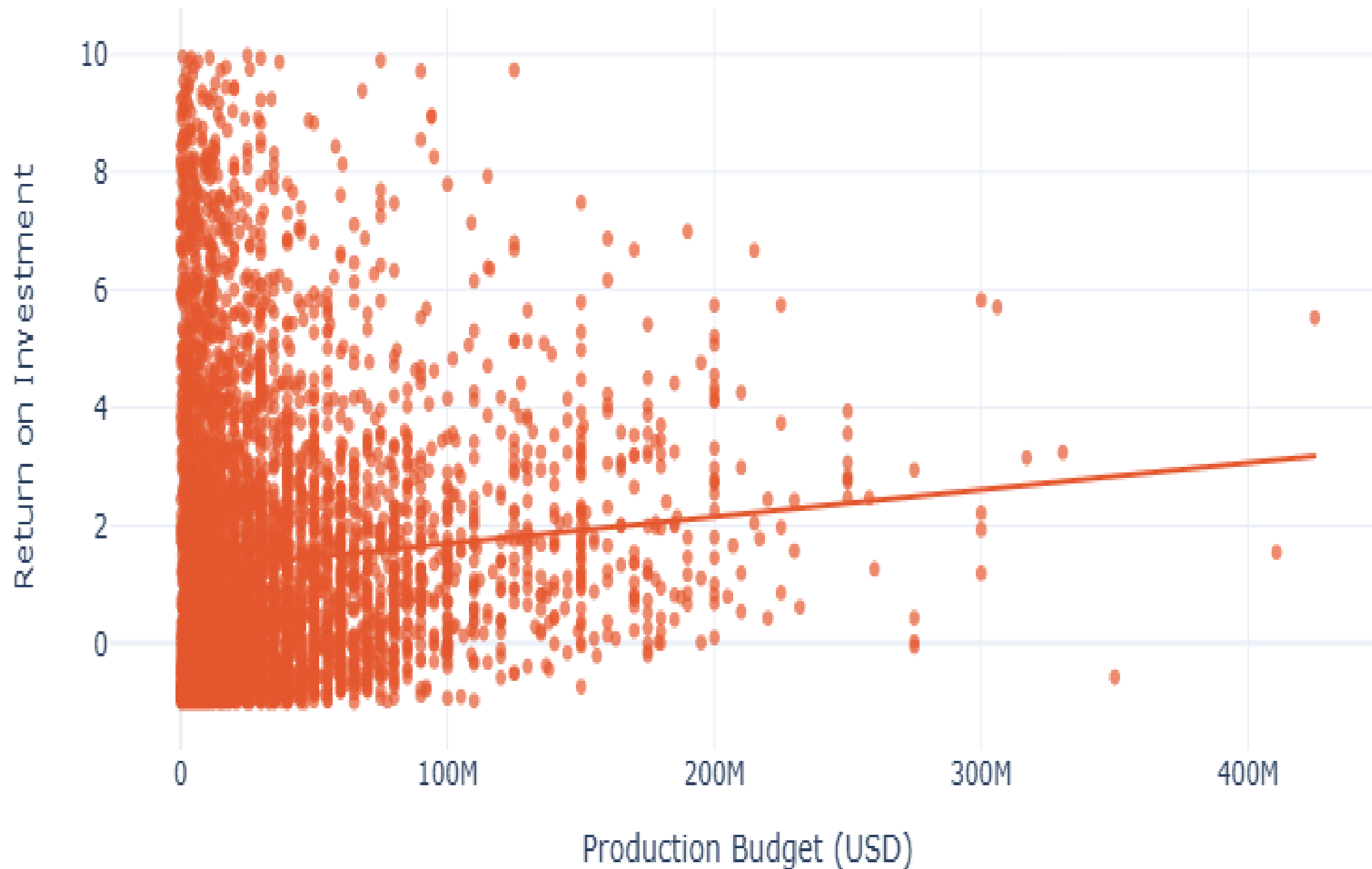
Observation

- **Overall Trend:** The Box Office revenue for all Maturity Ratings has increased significantly since the late 1980s, peaking in the 2000s and early 2010s, before declining in the later years
- **Dominance of PG-13:** The PG-13 rating consistently accounts for the highest box office revenue throughout the period from mid 2000s to the end of the chart, with revenue reaching over \$50 billion around 2010
- **Decline of G and NR:** The revenue for G (general audience) and NR (Not Rated) films remains consistently low throughout the entire period, rarely exceeding a few billion dollars and often staying close to zero



**WHAT BUDGET
RANGE YIELDS
THE BEST ROI?**

Production Budget vs ROI



- **Big Budgets $\neq$ Big Profits**

Spending more gets you bigger box office numbers, but not **better returns**.

Half of all films barely make back their budget.

25% lose over 30% of their investment.

- **Averages Are Misleading**

The "average" movie seems profitable (ROI $\sim 4x$), but that's skewed by rare mega-hits (*Avatar*, *Avengers*).

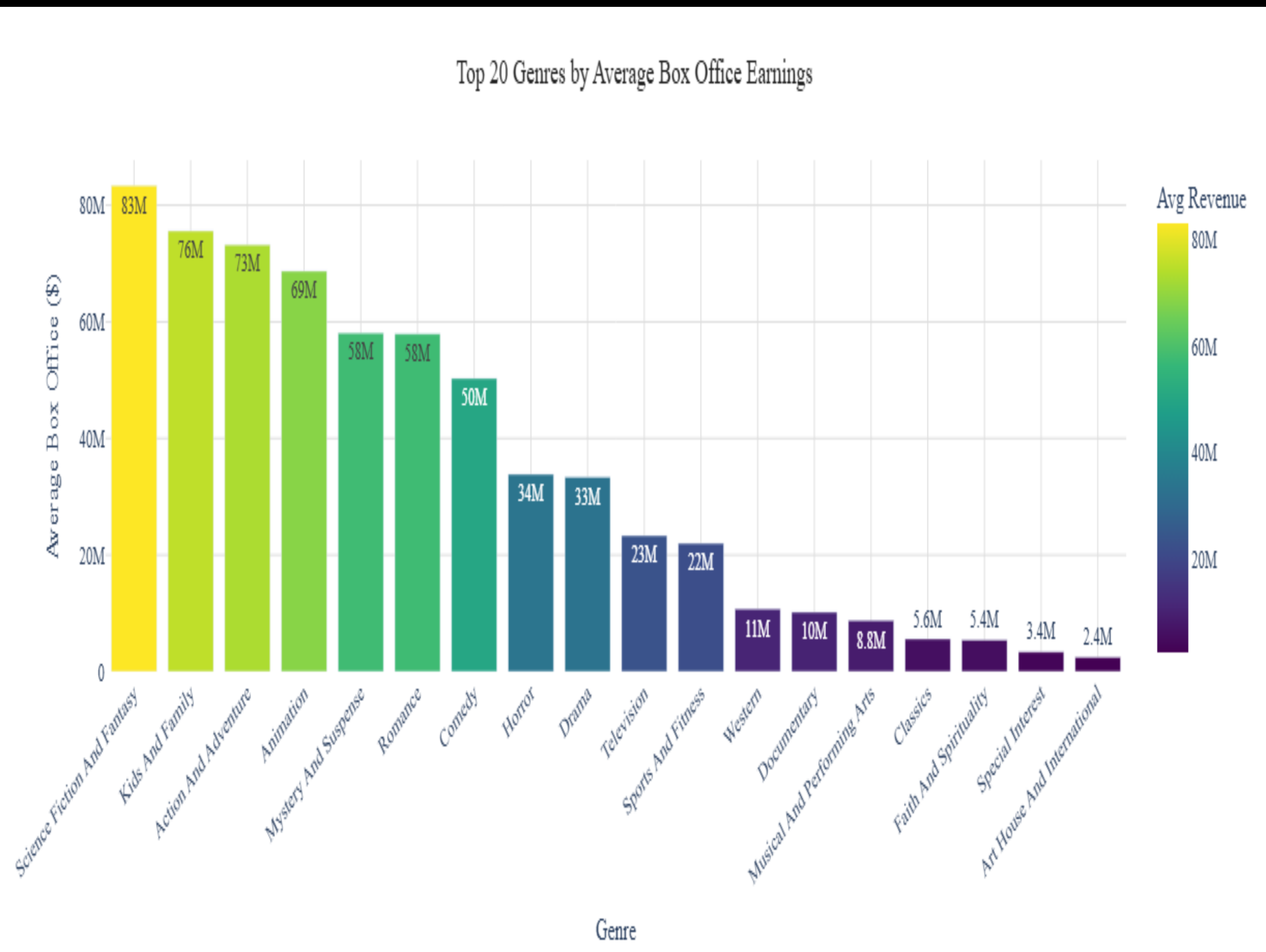
Most films (median ROI = 0.88x) lose money.

- **Small Budgets Punch Harder**

Films under \$20M often deliver **higher ROI** than blockbusters.

Examples: *Paranormal Activity*, *Get Out*, *The Blair Witch Project*.

WHICH GENRES ARE CONSISTENTLY EARNING THE HIGHEST REVENUE?



Observation

Top Dogs:

- Science Fiction & Fantasy leads the pack with the highest average returns—over \$83M.
- Kids & Family, Action & Adventure, and Animation follow closely, all pulling in \$68M—\$75M ranges.

Mid-Tier Successes:

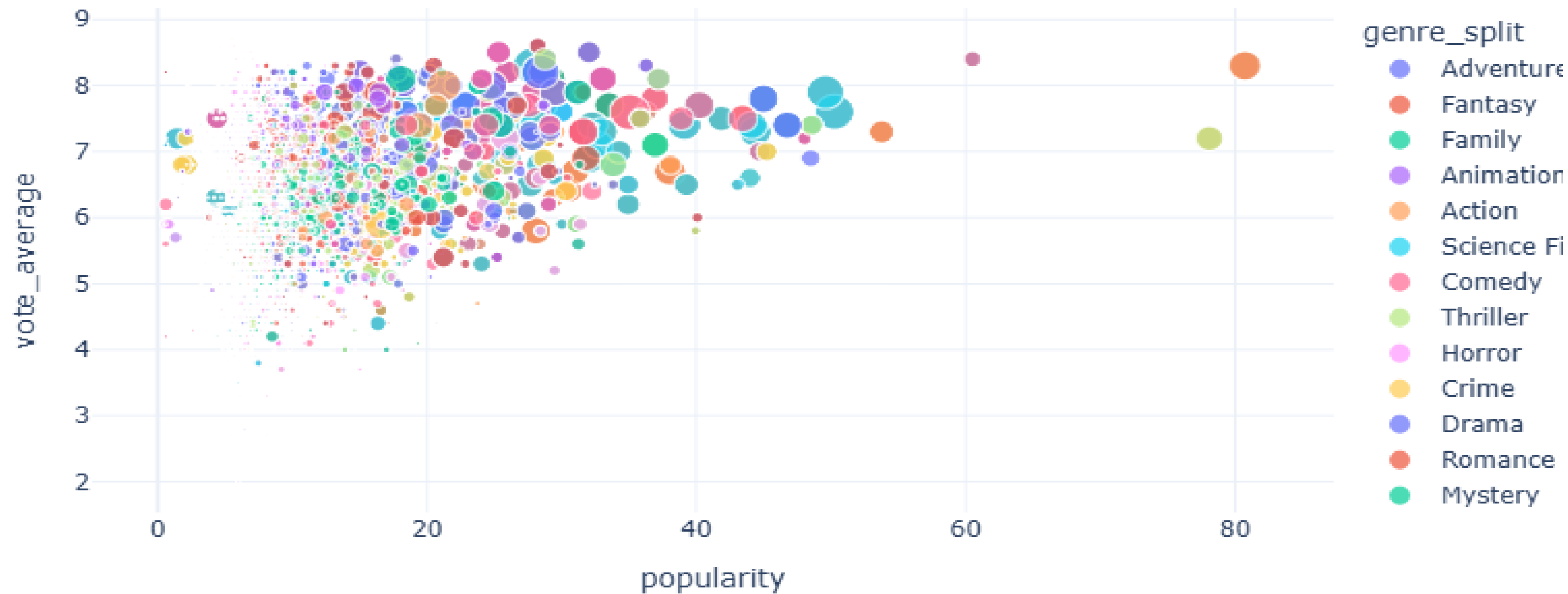
- Mystery & Suspense, Romance, and Comedy hover around the \$50M–\$58M mark. While they’re less spectacle-driven, they benefit from broad relatability, charismatic leads, and cultural timing

Niche and Specialist Fare:

- Genres like Documentaries, Musicals, Faith-based films, Westerns, and Art House see the lowest box office earnings—mostly under \ \$10M.

WHICH GENRES ARE CONSISTENTLY EARNING THE HIGHEST REVENUE?

Genre Popularity against Vote Average by Genre



- At first glance, we see a dense cloud of titles tightly packed between popularity scores of 0–30 and vote averages of 5–8. This is the heartland of cinema—where most films live, breathe, and compete.
- But what’s more telling are the outliers:
 - - A Romance film surges past 80 in popularity, scoring high in votes, showing that the genre can occasionally captivate both the masses and critics.
 - - A Thriller trails closely behind, also claiming a spotlight spot with exceptional popularity and decent ratings.
 - - Science Fiction and Drama genres cluster around the 7.5+ vote average - these genres seem to resonate well with audiences on quality, even if they don’t always dominate popularity.
- Meanwhile, genres like Documentary, TV Movies, and War appear more modest, with generally smaller bubbles and less extreme values - suggesting niche but loyal audiences.

HYPOTHESIS TESTING



Hypothesis 1

Research Question:

Does the maturity rating of a film (G, PG, PG-13, R) significantly affect its box office performance?

Null Hypothesis (H0):

There is **no difference** in the mean box office revenue across different maturity ratings.

Alternative Hypothesis (H1):

At least **one** maturity rating has a **significantly different** mean box office revenue.



Test Chosen : One-Way ANOVA

This test is appropriate because:

The independent variable (maturity rating) is categorical with 4 levels.

The dependent variable (box-office) is continuous.

We are comparing more than two group means simultaneously.

Assumptions:

1. Independent groups: Ratings are independent.

2. Normality: Each group's box office earnings should be roughly normally distributed.

3. Homogeneity of variances: Variance of earnings across groups should be similar.

Decision Rule:

- If $p\text{-value} < 0.05$, reject the null hypothesis. make this brief

Interpretation:

- $F(3, 14107) = 384.31, p < 0.001$, thus the result is highly statistically significant and we can confidently reject the null hypothesis.
- The extremely low p-value (< 0.0001) confirms that differences in box office revenue across maturity ratings are not due to chance.
- Therefore, **Maturity Rating has a statistically effect on box office revenue**

Strategic Insight for Ahjin Studios:

- ❖ **PG-13 and PG films** outperform **G and R-rated** films by **over 2x** in average revenue.
- ❖ For a new studio without franchise leverage or global name recognition, targeting a **PG or PG-13 rating** offers the **broadest demographic reach** (including teens and families) and **highest ROI Potential**.
- ❖ Avoid debuting with **R-rated content** unless aiming for niche prestige or festival circuits; the market signals lower revenue ceiling.
- ❖ **G-rated** films, while safe, may be **overly limiting** unless operating in the animation/kids content niche.

Recommendation:

- ❖ Ahjin Studios should **prioritize greenlighting PG or PG-13 rated films** for theatrical releases to maximize commercial success, especially in its initial production phase.
- ❖ These maturity brackets not only command the highest mean revenues but also align best with the global box office's appetite for broad-access storytelling.



HYPOTHESIS 2

Business Rationale:

- ❑ Ahjin Studios does not have the financial leverage to compete with \$100M+ blockbuster productions. To de-risk its market entry, it must identify the budget tier that offers the best trade-off between investment and return.

This hypothesis evaluates whether:

- ❑ **Mid-budget films**(\$10M–\$30M) deliver **significantly higher ROI** than their high-budget counterparts (>\$70M), making them a strategic sweet spot.

Null Hypothesis (H0):

- ❑ There is **no significant difference** in ROI between mid-budget and high-budget films.

Alternative Hypothesis (H1):

- ❑ **Mid-budget films** have a **significantly higher** mean ROI than high-budget films.

Test Chosen: Independent Samples T-Test (One-Tailed)

Groups:

- -Mid-Budget(production_budget between \$10M & \$30M)
- -High-Budget(production_budget above \$70M)

Dependent Variable: ROI

Test Type: One-tailed t-test to detect whether ROI is statistically greater in the mid-budget group.

Decision Rule:

- If **p-value < 0.05**,reject H0 - conclude that mid-budget films perform better.
- If **p-value $\geq$ 0.05**, fail to reject H0 – no evidence mid-budget films are more profitable.

Interpretation

A linear regression model was fit to assess how budget tier influences return on investment (ROI).

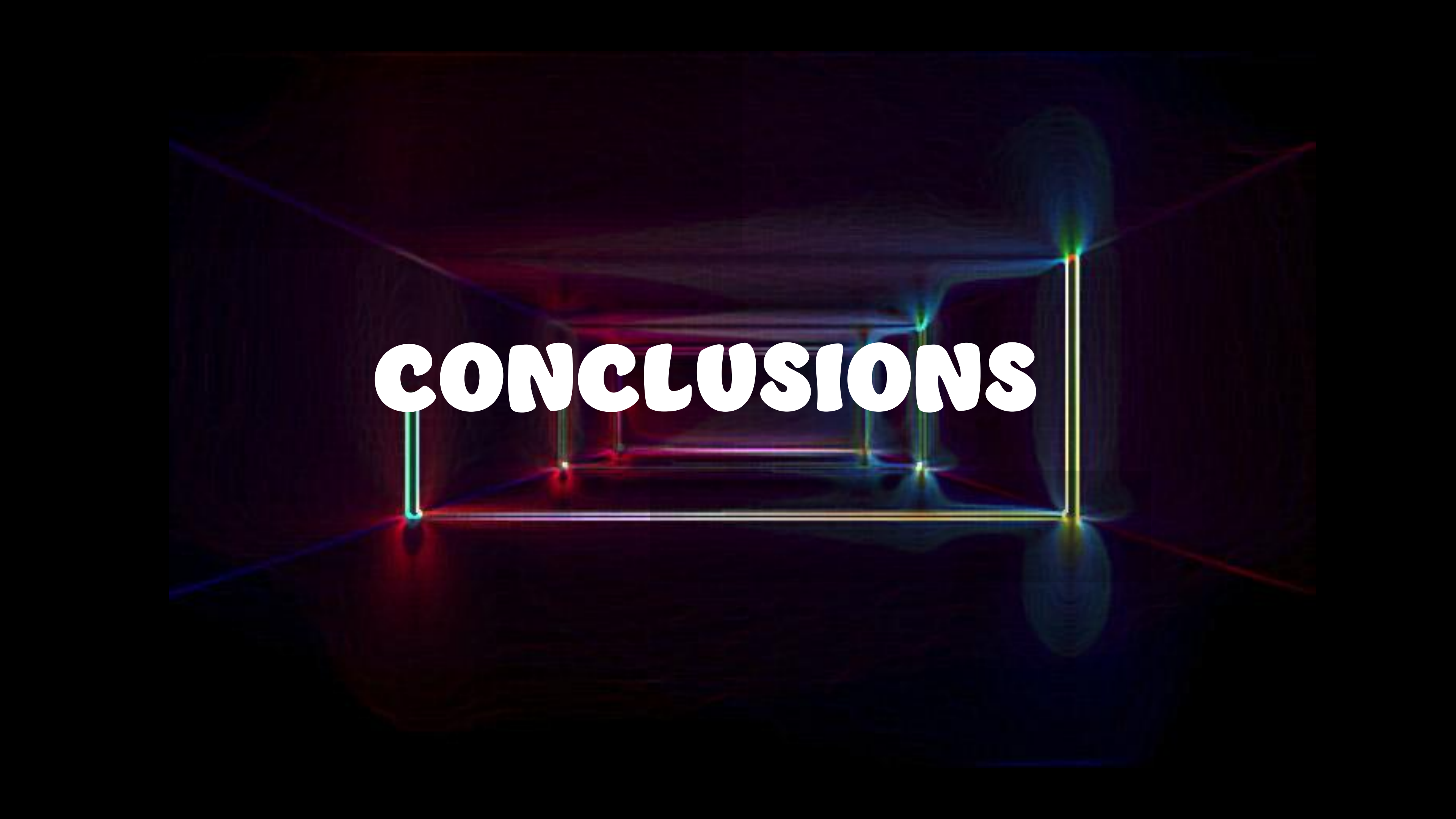
$R^2 = 0.008$ - While the model explains a small proportion of ROI variance, the difference in returns between tiers is still statistically meaningful.

What this means:

- Mid-budget films do not statistically outperform high-budget films in ROI. With a p-value of 0.649, there is no significant evidence to support the hypothesis.
- Low-budget films deliver significantly better ROI, reinforcing the high-risk/high-reward potential of lean productions.
- This finding suggests that profitability in film is less about spend, and more about efficiency, audience targeting, and potentially distribution strategy.

Recommendation

- Prioritize low-budget films (< \$10M) to maximize capital efficiency and minimize downside risk. Strong creative execution can still yield exceptional ROI.
- Re-evaluate the assumption that mid-budget films are a safer bet - the data doesn't support this claim. Without strong differentiators (e.g. IP, genre innovation), mid-budget investments may underperform.

The background is a dark, deep blue or black space. It is filled with numerous thin, glowing lines of light in various colors, including red, green, blue, and yellow. These lines appear to be part of a complex, three-dimensional grid or a series of intersecting planes. The lines are most concentrated in the center and fade out towards the edges. There are also some larger, more diffuse areas of light, possibly representing distant stars or nebulae. The overall effect is one of a vast, mysterious, and technologically advanced environment.

CONCLUSIONS

Strategic Summary for Ahjin Studios

1. Release Timing Matters

- - Nov-Dec : Prime window for major releases (holiday audiences, awards consideration).
- -Deploy mid-budget or high-concept films where competition is thinner but attention is available - especially for thrillers, dramas, or genre hybrids.

2. Runtime = Perceived Value

- - 150-180 min films (PG/PG-13) perform best—seen as "event cinema."
- - Avoid <90 min unless stylistically intentional (risk of seeming low-value).

3. Target PG/PG-13 for Commercial Viability

- - Broadest audience appeal (family-friendly, international markets).
- - Reserve R-rated for prestige/awards plays only.

4. Genre Strategy: Blend for Impact

- - High-ROI genres :Thrillers, mysteries, dramas (lower cost, strong returns).
- - Avoid VFX-heavy blockbusters until scale justifies risk.
- - Hybrid narratives (e.g., thriller + social commentary) maximize engagement.

5. Budget with Precision, Not Excess

- - Big budgets ≠ guaranteed profits(many flops are overfunded).
- - Tiered funding: Match budgets to genre, talent, and market potential.
- - Data-driven greenlights: Assess ROI potential early (script stage).



RECOMMENDATIONS

RECOMMENDATIONS

In Summary:

1. Build around PG/PG-13 narratives with emotional scope
2. Lean into genre smart , budget-disciplined storytelling
3. Let runtime, tone and genre be strategic signals, not creative afterthoughts





THANK YOU!